

VisionTag: QR code-Based Clothing Information System for People with Visual Impairments

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Abstract—Fashion reflects personality through flair and creativity, with color having the most impact. However, people with visual impairments often struggle to select clothing without relying on others, such as store staff or friends. Even after shopping, matching outfits becomes a challenge due to the vast range of color shades.

To address this, we propose VisionTag, an app designed to assist visually impaired individuals with offline shopping and wardrobe management. Using QR codes, users can scan garments and receive text-to-speech descriptions of properties including color, material, and texture. The application features gesture-based navigation with multi-modal feedback through audio, haptic, and visual responses.

Expert evaluation using Nielsen’s usability heuristics across four core tasks demonstrated that VisionTag successfully implements accessibility best practices, with only minor improvement opportunities identified (highest severity 3/9). Results confirm the application provides effective, accessible functionality for independent clothing management.

VisionTag also provides digital wardrobe management and offline shopping assistance, benefiting users both with and without visual impairments.

Index Terms—accessibility, visual impairment, clothing, QR code, mobile applications, assistive technology

I. INTRODUCTION

Clothing is a basic human need, yet it also functions as a medium for self-expression, reflecting personal identity, social status, and cultural norms [1]. Through color, texture, and design, individuals convey their personality and sense of style. However, for people with visual impairments, selecting appropriate clothing both while shopping and during daily wardrobe use can be a significant challenge due to the inability to access visual information.

Visually impaired individuals often rely on tactile markers, memory-based systems, or assistance from others to manage clothing information. Tactile systems such as Braille labels, custom stitching, or distinctive buttons are limited in the type and amount of information they can convey and may damage clothing over time [2], [3]. Memory-based strategies, while low-cost, introduce cognitive load and are prone to errors [2]. Assistance from sighted individuals either friends, family members, or store staff compromises independence and privacy [3], [4].

Technological solutions have emerged in recent years to address these issues. Tools like Envision AI and Microsoft Seeing AI provide general object and text recognition capabilities, but they lack specific features for clothing identification and typically require an active internet connection [5]. Devices like EyeRing [6] or color identifiers can detect basic color properties, but are often unable to interpret complex patterns or shades and may require specialized hardware [3], [6]. Applications such as Be My Eyes and VizWiz use crowd-sourcing approaches to bridge visual gaps [7], [6], but these introduce dependency and potential latency in real-time contexts.

Research has shown that users prefer objective and clear descriptions of clothing attributes, especially color and texture, rather than subjective opinions [4]. Additionally, clothing management systems should be easy to use, minimize reliance

on memory, and not require advanced literacy or assistive hardware [8], [9].

This paper introduces VisionTag, a mobile application that addresses the limitations of existing systems by providing accessible, detailed, and offline garment information using QR codes. The system delivers text-to-speech descriptions of color, material, and texture, supporting wardrobe organization and independent shopping for visually impaired individuals. By eliminating reliance on specialized hardware or internet access, VisionTag offers a practical, scalable solution to enhance autonomy and accessibility in clothing management.

II. RELATED WORK

Even today people with visual impairment still have problems with clothing management and shopping. Therefore different solutions have been investigated to help them till today and research is still continuing. We can put these related works into different groups.

A. Tactile Labeling Systems

Tactile tagging systems utilize physical markers to share information through touch. The “iro-pochi” system [10] uses tactile dots placed in a circle. These tactile dots represent different colors. So, they allow blind users to identify clothing colors by feeling the position of a raised dot or hole. Therefore, this approach requires specialized manufacturing. Also, it provides only limited information beyond the color. Similarly, Williams et al. [2] explored various tagging methods. Their analysis includes distinctive button shapes, pattern sewn onto the front of the shirt and RFID tags. Their results indicated that while simple tactile markers are easy to locate, they provide insufficient information. As a result, they put too much memory demands on users.

B. Crowd-sourcing Approaches

Applications such as VizWiz [11] use crowd-sourcing to answer visual questions. This means the application uses several people with sight to answer questions, including clothing identification. A quote from the paper states: “We hope that by building services like VizWiz which enable blind people to ask questions that are not yet possible to answer with automated approaches, we might help motivate research in supporting the types of questions that blind people actually want answered” [11]. In this sentence they comment that automated approaches will be better rather than using crowd-sourcing. Similarly, Be My Eyes connects users with sighted people who can help identify clothes [7]. However, recently in 2023, they also integrated the GPT-4 model into their app to make it more automated [12].

C. QR Codes, Barcode Scanners and Assistive Devices

EyeRing project [6] explores finger-worn assistive devices for recognizing colors and text. However, it requires a piece of special equipment to work. Therefore, it is not really eligible for everyone and requires users to learn and adapt to the use of this new device. On the other hand, Zappar’s “Accessible QR”

solution [13] enhances standard QR codes with a distinctive dot-dot-dash pattern. It makes them scannable from greater distances. Therefore, it improves the accessibility of this approach. This approach has the potential to modify QR technology to help visually impaired users with cloth management. Even though, this solution is implemented for packaged goods rather than clothing applications. [14] offers "Eu Visto" (I Wear), an app that assists visually impaired users in selecting suitable outfits by scanning QR codes on garments. Their app retrieves information about type, size, texture, and color. The system also uses NFC tags and an Arduino development board for clothing matching. Their approach received high usability ratings. However, users found the buzzer feedback difficult to interpret. On the other hand, Envision AI and Microsoft Seeing AI introduce general object recognition and barcode scanning features [5]. However, they do not offer any specialized readers for clothing tags. They also need an internet connection to access product information.

D. Information Accessibility Challenges

Research by van Staden et al. [8] and van der Merwe et al. [9] highlights that even sighted consumers with literacy challenges struggle to effectively use clothing label information. These studies highlight the need for a clear and accessible information display. They reveal that clothing displays should not rely on advanced literacy or visual abilities.

[4] found that blind people need help to shop for clothes. Color is most important for how they express themselves. Many feel uncomfortable asking store workers for help. This makes their disability more obvious. People in the study liked clear descriptions about colors. They preferred this over personal opinions. They also said texture matters when choosing clothes. This research shows clothing information should be easy to understand. It should not need good eyesight or help from others. This helps people feel more independent.

E. Limitations of Existing Approaches

Current solutions face several limitations:

- High costs (specialized hardware readers)
- Limited information storage (simple tags)
- Cognitive burden (remembering tag meanings)
- Dependence on others for initial setup

Our project builds on these findings and presents an alternative approach. By utilizing QR codes, we eliminate the high costs associated with implementation. Since we developed our application for mobile devices, there is not any issues with dependency. Furthermore, we also remove cognitive burden from user by simply storing data of clothes in their device's local storage.

III. DESIGN

We started by making medium-level prototypes on a tablet, as shown in Fig. 1, 2, and 3. These drawings showed our initial design ideas using simple tap interactions. They included screen layouts, basic navigation, and how information would

be organized. Using a tablet to sketch helped us test different ideas quickly and plan the overall user experience.

The wireframes show three main user flow scenarios from our initial tap-based design. Figure 1 demonstrates the main navigation and home mode functionality, including how users view their wardrobe items, access detailed item information, and update clothing status through tap interactions. Figure 2 shows two key home mode workflows: the remove item process where users can delete clothing items from their wardrobe, and the scan item functionality where users scan QR codes with the camera to get detailed item information. Figure 3 illustrates the complete retail mode functionality, showing both main retail workflows - scanning existing QR codes to add items to the wardrobe, and creating new QR codes by filling out detailed clothing information forms, with the final step showing the generated QR code that can be shared. Each wireframe includes arrows that show the user's journey between screens and notes that explain what happens at each step.

These hand-drawn prototypes represented our legacy design that relied on exploratory tapping, where users had to tap around the screen to discover functions. After testing this approach, we realized it was not ideal for visually impaired users. This led us to develop the new gesture-based navigation system with swipe controls and multi-modal feedback that we described in the following sections. The evolutionary development process helped us move from this initial tap-based design to a more accessible gesture-based interface.

In addition to a simple layout and interaction mode, feedback is also an essential component in producing accessible applications, thus we utilize multiple forms of feedback within our application particularly oriented towards visually impaired people. Modes of feedback include audio, haptic feedback and visual. These are mostly used in unison, to alert the user if an action is performed, guide the user through the application by reading out the current system status or provide contextual help. For instance, when switching between tiles on a page, we use slight haptic vibration to alert the user of their action, audio feedback of the current selected tile name is provided along with contextual help if applicable, selected tile is highlighted which might be a slight convenience for visually impaired people, but as stated we aim to make our application inclusive to visually impaired people along with those without visual impairment.

The following section provides a detailed explanation of each page in our application breaking down the design iterations it went through. We've labeled the earlier versions as Legacy Design and the final versions as New Design, to visually compare evolution of each page. Alongside these comparisons, we've explained why certain changes were made. We also detail the interaction modes we tested, the variations we explored, and the pros and cons that influenced our decisions.

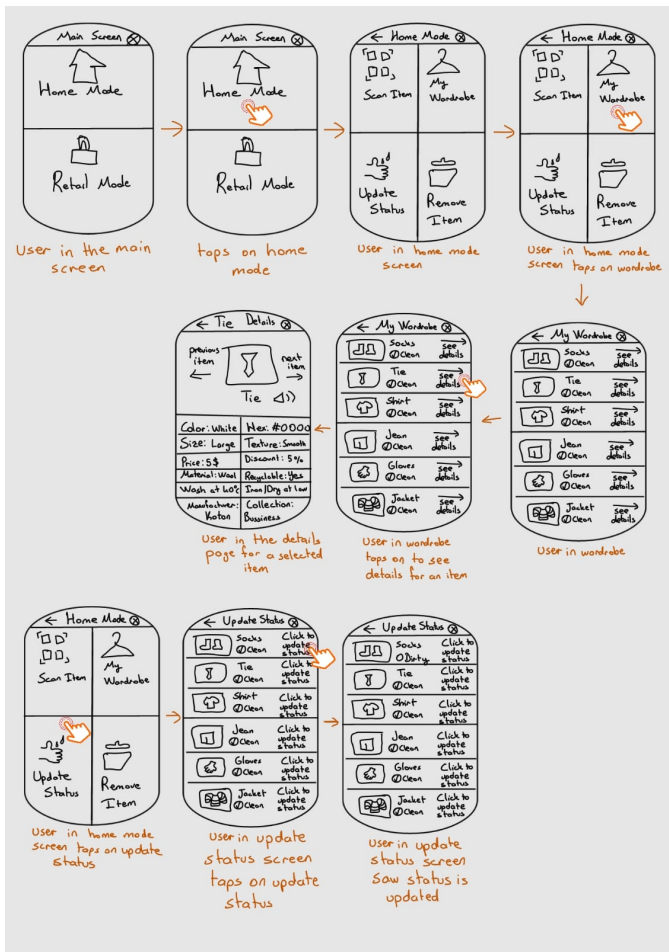


Fig. 1. Initial wireframes showing main navigation and home mode functionality

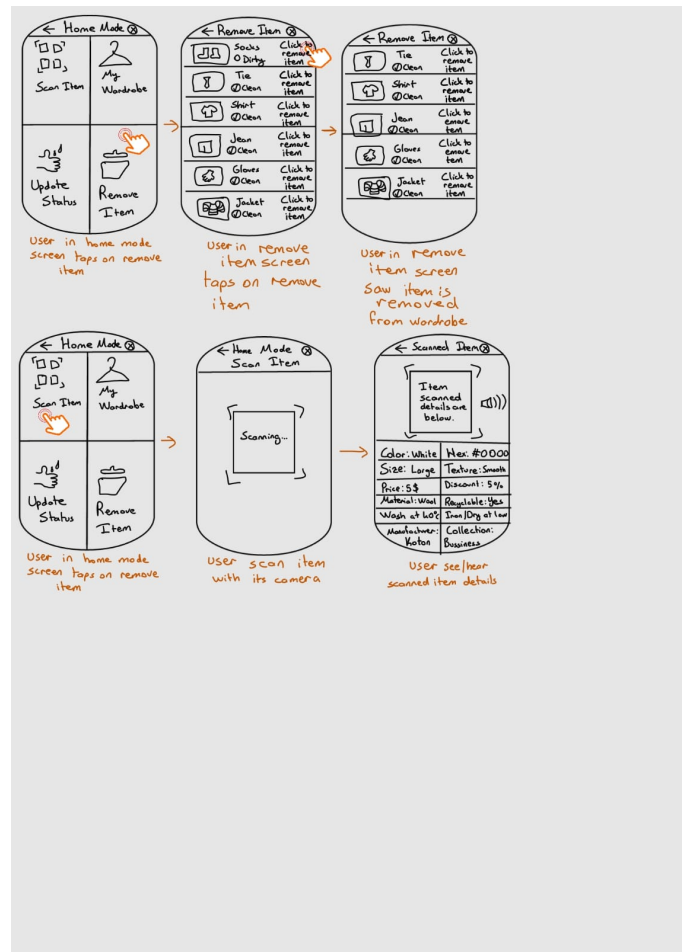


Fig. 2. Initial wireframes showing key home mode workflows

A. Main Screen and Navigation

We follow one general rule for navigation across all pages in the app: a consistent set of interaction modes for moving through and interacting with content. This has gone through several iterations to find what feels most intuitive. In the legacy design, we used an exploratory approach where the user taps anywhere on the screen and the system reads out whatever element they touched, while providing feedback about what they tapped, but doesn't select or activate. With only double tap on specific item to activate it.

Legacy Design - Challenges

Users have to tap around the entire screen to discover all available functions, and even long-time users must remember the location of each element. Additionally, tapping between buttons often results in mistakes. To fix this, we introduced audio prompts that list the available functions on each page, along with positional cues like "top left" or "bottom center" to help users build a mental map. Still, this was far from ideal as screen layouts vary across the app, and orientation remains

a challenge.

New Design

We've moved to a gesture-based navigation model paired with tap-to-activate functionality. Each page offers at least two functions, with one automatically selected when the page loads this is confirmed via voice feedback and a visual highlight. Users can swipe down to cycle through other functions, and double-tap to activate the selected one, as shown in Fig. 4.

New Design - Benefits

Functions are easier to locate, fewer mistakes are made as only the selected item responds to double-tap, even if the user taps elsewhere. Because of these improvements, this interaction model forms the core of our current design and can be seen across all pages within the application.

B. Home Mode

Home Mode offers four core functions:

- Wardrobe - view items in the wardrobe
- Scan - add new items to the wardrobe

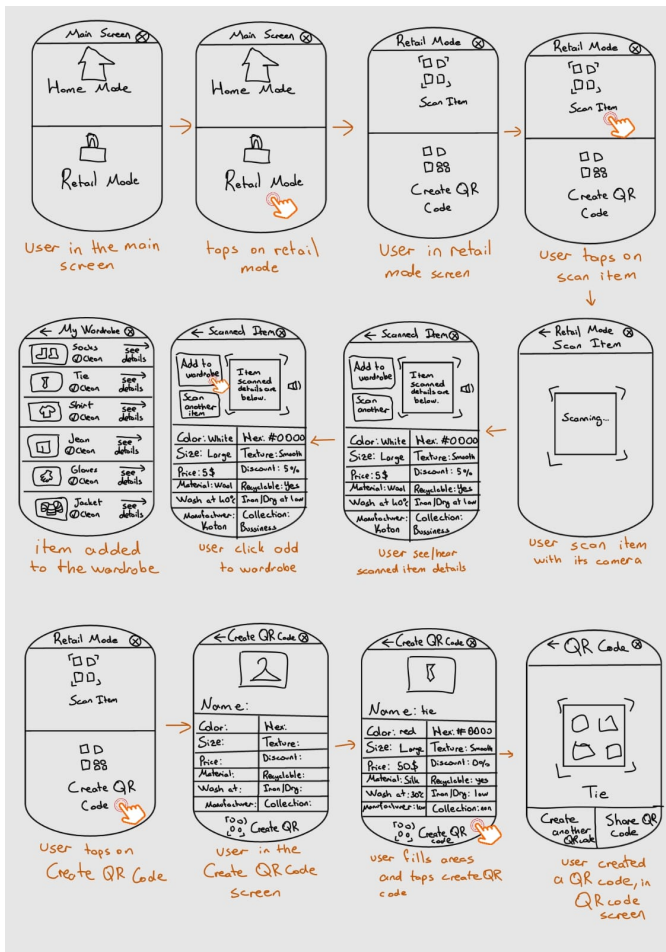


Fig. 3. Initial wireframes showing complete retail mode functionality

- Update - modify item status (e.g. worn, washed)
- Remove - delete items from the wardrobe

In the legacy design, we envisioned these functionalities being directly accessible from Home Mode via four corner tiles each representing one function. This made all actions immediately available, with no need to remember steps or navigate deeper into menus. The user also had a clear understanding of what could and couldn't be done within Home Mode.

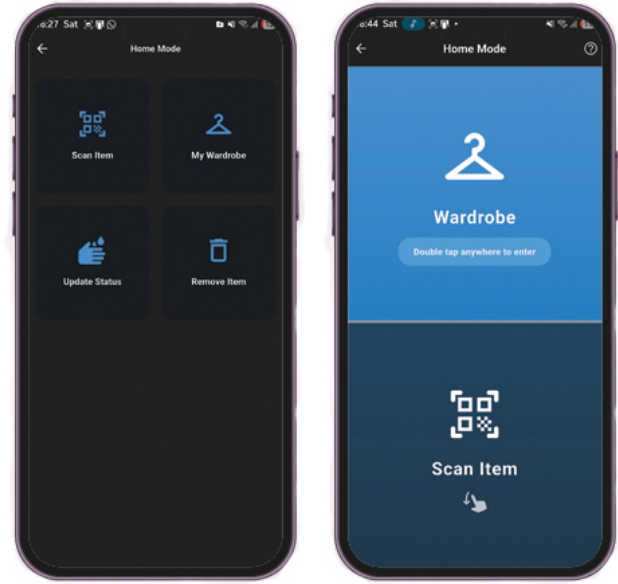


Fig. 5. Home Mode Legacy(left) vs New (right)

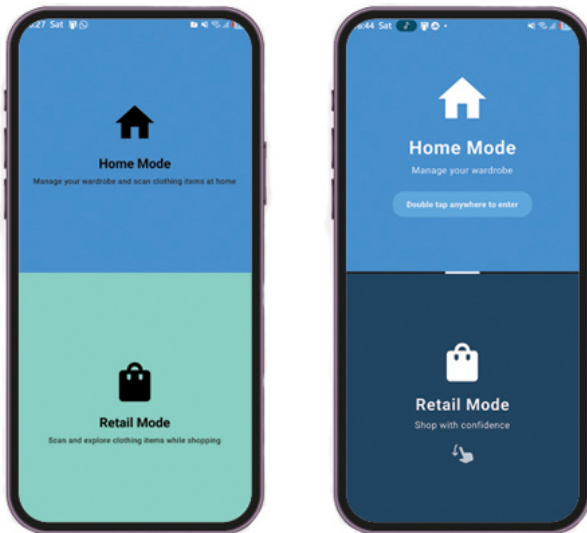


Fig. 4. Home Legacy(left) vs New (right)

Legacy Design - Challenges

TTS (Text-to-Speech) had to read out all four functions upon entry, which overwhelmed users. The screen felt cluttered, especially since Update and Remove function are more like sub-actions under Wardrobe. Additionally, interaction required precise tapping, often leading to errors.

New Design

We restructured the layout by grouping Update and Remove under Wardrobe, reducing the number of top-level actions to just two: Wardrobe and Scan. This allowed us to adopt the new gesture-based navigation model. When the user enters Home Mode: Wardrobe is selected by default and visually highlighted on entry. Swiping down switches the selection to Scan, and a double-tap anywhere on the screen activates the currently selected function.

New Design - Benefits

There's no need for precise taps since only the selected function responds, which reduces errors. Users aren't overwhelmed by multiple options at once, making navigation more

intuitive. Swiping provides a faster and easier way to move through options compared to exploratory tapping. This updated interaction model makes Home Mode less error-prone, and easier to use for both new and experienced users.

C. Wardrobe

The Wardrobe mode allows users to view their items and check properties like name, color, size, and status. In the initial version shown in Fig 6 on the Left, we presented items in a scrollable list. Users could tap an item to hear a brief description, and double-tap to open full details.

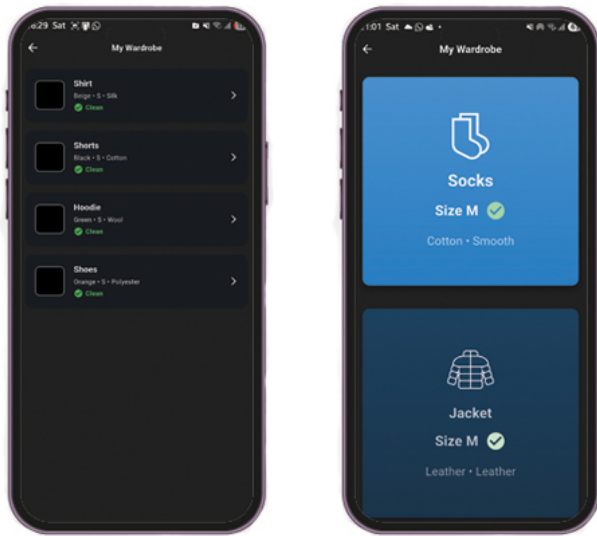


Fig. 6. Wardrobe Mode Legacy(left) vs New (right)

Legacy Design - Benefits

All items were accessible from a single page, and the list format aligns with real-world expectations, such as scrolling through a catalog.

Legacy Design - Challenges

Having too many items crowded the screen made it difficult for users to focus. They often had to remember exactly where they last tapped to find a specific item. Additionally, the "read out" button was small and located at the bottom of the screen, making it especially hard for visually impaired users to find.

New Design

We redesigned the layout to show only two items per page, as shown in Fig. 6 on the Right. Following the same interaction logic as Home Mode, one item is selected and highlighted by default. Users can swipe to switch between items and double-tap to access more information.

For additional wardrobe actions Update and Remove, we built in gestures directly on the selected item:

- Tap and hold to change the item's status
- Pinch out/Zoom in to delete the item

New Design - Benefits

Interacting with items becomes easier and more focused with only two items displayed per screen, reducing cognitive and visual load. Using gestures instead of extra buttons further improves accessibility and creates a smoother interaction flow.

D. Retail Mode

The legacy design of Retail Mode actually helped shape the new interaction model. From the beginning, this mode only had two core functions—Scan and Create QR Code—so it felt intuitive to represent them as just two buttons. The goal was a clean, minimal interface that felt simple and easy on the eye. Legacy design is shown on the left side of Fig. 7 on the Left.

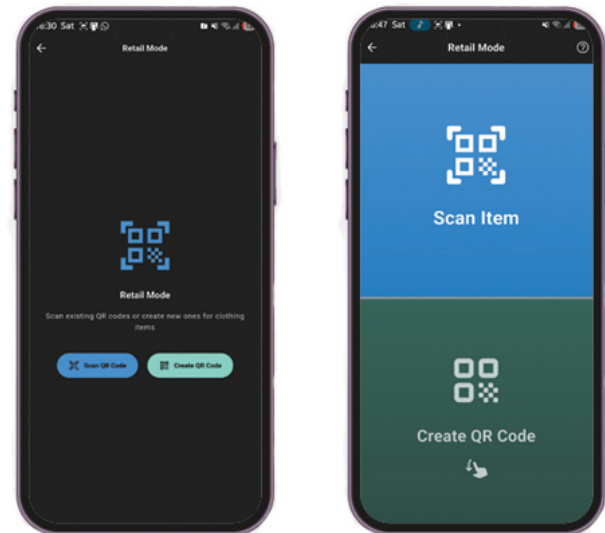


Fig. 7. Retail Mode Legacy(left) vs New (right)

Legacy Design - Benefits:

The layout is minimalistic, providing direct tap access without requiring any gestures.

Legacy Design - Challenges

The buttons were too small, which limited the interaction area and made tapping difficult. Additionally, the layout design was inconsistent with other modes, contributing to confusion. This combination increased the likelihood of errors, especially when users accidentally tapped between buttons.

New Design

To maintain a consistent layout across the app, we applied the same two-item navigation model used in Home Mode, give in Fig. 7 on the Right. This leverages the new interaction design:

One option is selected and also providing haptic feedback, allowing the user to immediately feel the selection change. Users can swipe to switch between options, and a double-tap anywhere on the screen activates the currently selected function.

New Design - Benefits

The consistent interaction and layout across app modes make it easier for users to navigate, as they can rely on their familiarity with other pages. Additionally, the tapping design doesn't require precise targeting, which helps reduce mistakes and improves overall usability.

E. Create Qr-Code

For this mode, the goal was to let users enter item details through a simple, structured form. In the legacy design shown in Fig. 8 (Left), we laid out all the input fields vertically from top to bottom on a single scrollable page. A "Generate QR Code" button was placed at the bottom.

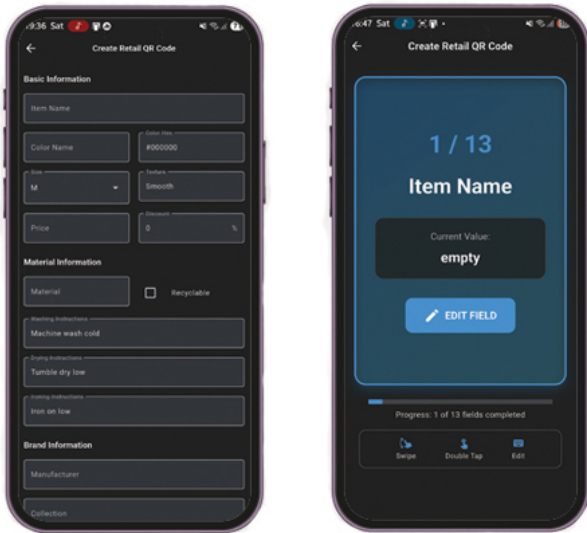


Fig. 8. Create Qr Code (left) vs New (right)

Legacy Design - Benefits

All required fields were visible at once Users could quickly review previously entered information and spot errors

Legacy Design - Challenges

Displaying too much information on a single screen made it difficult for users to focus, especially those with fixation

issues, as seeing multiple field labels at once could lead to confusion or input errors. Additionally, the "Generate" button was small and hard to reach, making it less accessible. Overall, the layout did not align with the navigation and interaction model used in other parts of the app, creating an inconsistent user experience.

New Design

To improve focus and maintain consistency with the rest of the app, we now display one field at a time. The user can double-tap to edit the current field and swipe to move to the next or previous one. A progress bar is also included to indicate how many fields remain, helping users track their progress throughout the form. Illustration of the new design is shown in Fig. 8 (Right).

New Design - Benefits

This approach reduces cognitive load by showing only one input field at a time, which helps improve focus and minimizes distractions. It also reinforces consistent navigation patterns such as swiping to navigate and tapping to edit which aligns with the overall interaction logic used across other modes in the app.

New Design - Challenges

Users don't have a clear overview of all the fields upfront, the entry order is semi-fixed which limits flexibility, and mistakes are harder to catch early on often only becoming apparent at the end of the process.

IV. IMPLEMENTATION

For implementation, we aimed to choose a development SDK for Android that was easy to learn, well-documented, and supported by a strong developer community. We selected Flutter [15], an open-source a cross-platform development framework that supports Android, iOS, Windows, and macOS. Flutter's flexibility allows us to develop for multiple platforms with minimal additional effort. It also offers a wide range of visually appealing UI components out of the box, making it easier to build a polished interface without much customization. Flutter uses the Dart programming language, which shares similarities with C++. Since both team members already had experience with C++, transitioning to Dart was relatively straightforward and required minimal overhead.

A. Text-to-Speech

Our application heavily relies on audio assistance and feedback to guide users through the interface and confirm their actions. To support this, we considered two approaches: standard Text-to-Speech (TTS) systems and deep learning-based voice synthesis, each with its own pros and cons.

TTS (Text-to-Speech):

- Natively supported on most devices through built-in accessibility features

- Requires minimal setup only helper functions to pass text to the OS-level TTS engine
- Lightweight in terms of memory usage and computational requirements
- Enables near real-time feedback

The synthesized voice often sounds robotic or less natural. In contrast, deep learning-based voice synthesis produces more natural, human-like speech. Although deep learning requires high memory and computational cost, while providing slower inference, resulting in noticeable latency making it less practical for real-time interaction on mobile devices.

To balance responsiveness and accessibility, we used `flutter_tts` [15] plugin for flutter, which interfaces with the Android system's native TTS engine, allowing us to achieve the following.

- Set the voice and language
- Feed in text dynamically
- Generate audio with minimal delay

Our focus was on responsiveness ensuring users receive instant audio feedback when interacting with the application. Standard TTS provides that reliability, making it the more suitable option for our use case.

B. QR Code Generation and Scanning

To support QR code scanning and generation, we used the Mobile Scanner [16] plugin for Flutter, available on `pub.dev`. This package provides a unified API that works across Android, iOS, macOS, and Web, making it a flexible and platform-inclusive solution for our application.

QR Code Scanning

When the user enters Scan Mode, the device's camera is automatically activated, and the app begins scanning for a QR code. During this process:

- The user receives real-time feedback as the app continues searching
- Once a QR code is detected, the app provides audio and visual confirmation
- The QR code is then decoded using the same package, and the extracted information is displayed on screen
- At this point, the user is prompted to take action using gestures:
 - Swipe up to initiate adding the item to the wardrobe
 - Swipe up again to confirm
 - Swipe down to cancel
 - Swipe right to re-scan

QR Code Generation

The same library is used to generate QR codes. The version of the QR code is automatically inferred based on our item data structure. The user is guided through a series of input fields (as defined in `tab`). Once completed, the entered data is encoded into a QR code. The resulting image can be viewed, edited, or shared with other devices. This approach ensures consistency

in handling both input and output of item data via QR codes, while keeping the interaction smooth and accessible.

C. Gestures

Our application takes advantage of the wide range of sensors available on most mobile devices to support intuitive, gesture-based interaction. Since we target touchscreen devices, we implemented a variety of single and multi-finger gestures, tap interactions, and device shake gestures to make navigation and functionality more accessible especially for users with visual impairments. Flutter natively supports gesture detection through plugins like `sensorsplus` [17], a Flutter plugin to access the accelerometer, gyroscope, magnetometer and barometer sensors. Using this tool, we detect actions performed on the screen (such as drag direction or number of fingers) and trigger appropriate responses. For example, dragging a finger from left to right moves to the next page, while a double tap opens the currently selected function. In addition to screen gestures, we implemented shake gestures using the `sensorsplus` package. This package gives us access to device sensors like the accelerometer and gyroscope, allowing us to detect motion. When the device is shaken, we analyze the movement's intensity and speed to determine if it qualifies as a shake event. This gesture is used to trigger context-specific help. Refer to Fig. 9 for all supported gestures.

Note: In our app, a swipe is interpreted as a drag gesture that moves between pages.

2 Finger Drag left → next page

2 Finger Drag right → previous page

Gestures Used

Single-Finger Gestures

- Single Tap - Read out information
- Double Tap - Open selected function (navigate to associated page)
- Hold Change item status (works on wardrobe items)
- Special: Double Tap - Show more item details (in wardrobe mode)

Single-Finger Swipes

- Swipe Right - Go to previous page
- Swipe Up / Down - Navigate between page tiles
- Special: Swipe Up - Exit mode (in camera/scan mode)

Two-Finger Swipes (Wardrobe mode only)

- Swipe Right - Previous page
- Swipe Left - Next page

Shake Gesture

- Light Shake - Read out hints

Pinch Gestures

- Pinch Out - Delete item from wardrobe
- Pinch In - Close the application (works globally)

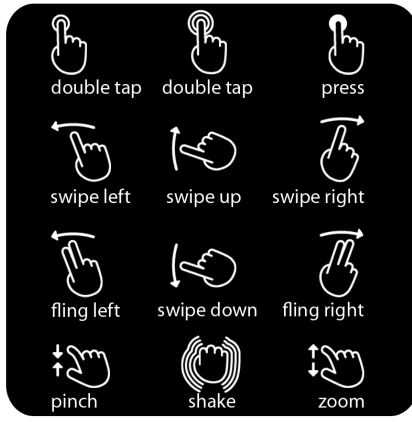


Fig. 9. Supported Gestures.

D. Storage

For storage, we opted for an offline-first solution that allows users to access their data without requiring an internet connection. Each wardrobe item is represented as a JSON data structure, which is lightweight, easy to parse, and can be managed natively within the app. An example of the clothing item structure is shown in 10. We store these JSON objects locally using Shared Preferences. This method is ideal for small datasets and is both lightweight and efficient for quick reads and writes. It also ensures that user data persists between sessions without needing external storage access or cloud connectivity. To manage the JSON data, we used Dart's `dart:io` and `dart:convert` libraries, which provide functionality for creating, modifying, and serializing JSON objects.

```
Clothing Item Json Object
|— id: string,
|— name: string,
|— color: string,
|— colorHex: hexadecimal,
|— size: string,
|— texture: string,
|— price: double,
|— discount: double,
|— material: string,
|— recyclable: boolean,
|— laundryInstructions: Map<String, String>,
|— manufacturer: string,
|— collection: string,
|— isClean: boolean,
```

Fig. 10. Clothing item json structure.

V. EXPERIMENTATION METHODOLOGY

For the expert evaluation, we used heuristic evaluation because it is an effective way to find usability problems in assistive technology applications. The evaluation was based on Nielsen's ten usability heuristics [18], with special focus on accessibility for visually impaired users. We tested each heuristic on the VisionTag mobile application while completing different tasks.

We focused on four main tasks that represent the complete functionality of VisionTag:

- **Task 1:** Moving through the application pages and onboarding screen using gestures and audio feedback.
- **Task 2:** Using Home Mode to manage clothing items in the wardrobe with gesture controls like selecting items, changing status, and deleting items.
- **Task 3:** Using Retail Mode to scan QR codes from clothing items with camera detection and audio confirmation.
- **Task 4:** Using Retail Mode to create new clothing QR codes through a form with 13 input fields.

We tested each heuristic on the VisionTag application while doing these four tasks to cover all major features and interactions. We rated the severity of problems on a 0-8 scale, where:

- 0 = Not a usability problem
- 1 = Very minor cosmetic issue
- 2 = Cosmetic problem only, low priority
- 3 = Minor usability problem, slight inconvenience
- 4 = Moderate usability problem, medium priority
- 5 = Significant usability problem, high priority
- 6 = Major usability problem, urgent attention needed
- 7 = Severe usability problem, critical functionality affected
- 8 = Usability catastrophe, system unusable

The evaluation process looked at the application's gesture-based navigation, text-to-speech integration, haptic feedback, and overall accessibility. We paid special attention to how well the multi-modal interaction design works for users with visual impairments, including audio announcements, vibration patterns, and simple gesture controls.

We completed the expert evaluation by going through each screen while doing the four tasks, checking all 10 heuristics and writing down areas for improvement. The evaluation took about 45 minutes to complete, covering all major user flows including initial navigation, onboarding tutorial, home mode wardrobe management, retail mode QR scanning, and QR code generation.

VI. EVALUATION AND RESULTS

The expert evaluation of the VisionTag mobile application showed good usability design using Nielsen's ten usability principles, with strong attention to accessibility for visually impaired users. The application follows established accessibility guidelines well and represents a good assistive technology solution.

A. Task-Specific Evaluation Results

Task 1: Application Navigation and Onboarding

The navigation and onboarding experience shows good design. The onboarding tutorial teaches users VisionTag's gesture vocabulary through hands-on learning. Each gesture is explained with clear audio followed by practice opportunities. The tutorial builds user confidence by making users actually perform gestures instead of just listening to instructions. The main navigation between Home Mode and Retail Mode provides clear audio announcements and swipe-based selection that is easy to understand.

Heuristics Successfully Supported: This task strongly supports Heuristic 1 (Visibility of System Status) through comprehensive audio announcements throughout the onboarding process. Heuristic 6 (Recognition Rather Than Recall) is well-implemented as the tutorial reduces memory load by providing immediate practice opportunities. Heuristic 10 (Help and Documentation) is effectively addressed through the step-by-step gesture learning system.

Heuristics Violated: No significant heuristic violations were identified for this task.

Task 2: Home Mode Wardrobe Management

The wardrobe management interface shows good accessibility design through its interaction system. Users can move between clothing items using single-finger swipes, with audio announcements giving item names, status, and available actions. The double-tap gesture for detailed item information provides complete accessibility descriptions including material, size, color, and care instructions. The long-press gesture for status changes (clean/dirty) gives immediate haptic and audio feedback. The zoom-based deletion system prevents accidental deletions while staying accessible through gesture controls.

Heuristics Successfully Supported: This task strongly supports Heuristic 1 (Visibility of System Status) through clear audio feedback for all actions. Heuristic 5 (Error Prevention) is well-implemented through the zoom-based deletion system that prevents accidental deletions. Heuristic 3 (User Control and Freedom) is supported by providing clear status change options and navigation controls.

Heuristics Violated: Minor violation of Heuristic 4 (Consistency and Standards) due to slight variations in navigation patterns compared to other parts of the application. For instance, in the Wardrobe section, a two-finger swipe left or right is used to navigate between pages, whereas elsewhere in the application, a single-finger swipe is used. This inconsistency may be confusing for users.

Task 3: Retail Mode QR Scanning

The QR code scanning functionality shows good integration of camera technology with accessibility features. The scanning interface provides audio guidance for camera positioning and gives feedback during scanning. The application detects QR codes successfully and provides immediate audio confirmation when codes are found. The scanning area overlay provides visual context while audio announcements help users understand scanning progress and completion.

The QR code scanning functionality works well overall, but we identified one minor area for improvement:

Heuristic Violated: Minor violations include Heuristic 10 (Help and Documentation) regarding progress feedback during QR scanning. The system should offer clearer instructions about which direction to move the camera and provide feedback about environmental conditions that may affect scanning success, such as lighting that is too dark or too bright. This would help users achieve successful scans more efficiently, particularly in challenging environments.

Heuristics Successfully Supported: This task strongly supports Heuristic 1 (Visibility of System Status) through continuous audio feedback during scanning operations. Heuristic 2 (Match Between System and Real World) is well-implemented as the scanning process follows familiar camera interaction patterns. Heuristic 9 (Help Users Recognize, Diagnose, and Recover from Errors) is supported through clear feedback when scanning fails or succeeds.

Task 4: Retail Mode QR Generation

The QR code creation system provides a working solution for complex form interaction through gesture controls. The 13-field form interface keeps accessibility throughout the completion process, with navigation between fields using vertical swipes. Each field type (text input, selection lists, toggles) uses appropriate interaction methods while providing audio feedback. The field completion progress is communicated, and the final QR generation process confirms successful completion.

Heuristics Successfully Supported: This task supports Heuristic 2 (Match Between System and Real World) through familiar form interaction patterns. Heuristic 3 (User Control and Freedom) is implemented through field navigation controls and the ability to modify entries. Heuristic 8 (Aesthetic and Minimalist Design) is supported by organizing complex information into manageable field-by-field interactions.

Heuristics Violated: Minor violations include Heuristic 1 (Visibility of System Status) regarding progress feedback during QR generation, Heuristic 4 (Consistency and Standards) due to navigation pattern variations, Heuristic 6 (Recognition Rather Than Recall) regarding field requirement reminders, and Heuristic 10 (Help and Documentation) regarding contextual guidance for complex workflows.

TABLE I
SUMMARY OF EXPERT EVALUATION FINDINGS

Improvement Opportunity	Violated Heuristic	Severity Rating	Related Task
Better progress feedback for QR generation	Heuristic 1: Visibility of System Status	2	Task 4
Minor gesture navigation standardization	Heuristic 4: Consistency and Standards	2	Tasks 2 & 4
More field requirement reminders in forms	Heuristic 6: Recognition Rather Than Recall	1	Task 4
Extended contextual help for complex workflows	Heuristic 10: Help and Documentation	3	Task 4
Camera guidance for QR code positioning	Heuristic 10: Help and Documentation	2	Task 3

B. Areas for Improvement and Severity Ratings of Findings

The application has strong overall usability, but we found a few minor areas where small improvements could make the user experience better:

Task 1: Application Navigation and Onboarding

The navigation and onboarding process works well overall. We did not identify any significant usability issues for this task. The gesture learning system and audio feedback provide effective guidance for new users.

Task 2: Home Mode Wardrobe Management

The wardrobe management system functions effectively. We found one minor area for improvement:

Heuristic 4 (Consistency and Standards) - Severity 2: The gesture vocabulary is used consistently, but there is a minor variation between the item navigation patterns in the wardrobe and the field navigation in the retail form. The vertical swipe gestures work well for item navigation, but could be made slightly more consistent with navigation patterns used elsewhere in the application. This minor inconsistency does not hurt functionality but represents an opportunity for better standardization.

Task 3: Retail Mode QR Scanning

The QR code scanning functionality works well overall, but we identified one minor area for improvement:

Heuristic 10 (Help and Documentation) - Severity 2: While the application provides audio guidance for camera positioning, there is an opportunity to provide more specific directional guidance to help users position the camera closer to QR codes. The system should offer clearer instructions about which direction to move the camera and provide feedback about environmental conditions that may affect scanning success, such as lighting that is too dark or too bright. This would help users achieve successful scans more efficiently, particularly in challenging environments.

Task 4: Retail Mode QR Generation

This task represents the most complex workflow in the application, and we identified several minor areas for improvement:

Heuristic 1 (Visibility of System Status) - Severity 2: The application provides good audio feedback throughout the form completion process, but there could be more detailed progress information during QR code generation. The current system announces the start and completion of QR generation, but additional progress updates (like "processing field validation" or "generating QR data") could give users better confidence during this multi-step operation.

Heuristic 4 (Consistency and Standards) - Severity 2: While the vertical swipe gestures work well for field navigation in the 13-field form, they could be made more consistent with the item navigation patterns used in wardrobe management (Task 2). This represents an opportunity for greater standardization across all application contexts.

Heuristic 6 (Recognition Rather Than Recall) - Severity 1: The application reduces memory load well through audio announcements, but there is a minor opportunity to improve the form experience by providing more frequent reminders about completion progress and required versus optional fields during the long form completion process. The current system announces field requirements well, but periodic progress summaries could reduce cognitive load for users completing complex forms.

Heuristic 10 (Help and Documentation) - Severity 3: While the help system through long-press gestures and shake-to-repeat functionality works well for basic navigation guidance, there is a moderate opportunity to improve contextual help specifically for the complex QR generation workflow. More detailed step-by-step guidance for first-time users working through the 13-field form could make this advanced functionality more accessible.

C. Summary of Expert Evaluation Findings

The expert evaluation found only 6 minor improvement opportunities in the VisionTag application across all four tasks, with the highest severity rating being 3 (minor usability problem). These findings show that VisionTag implements accessibility best practices and provides an effective, usable interface for visually impaired users across all major functionality areas. Tasks 1 and 3 showed no significant usability issues, Task 2 had one minor consistency issue, while most identified opportunities relate to Task 4 (QR generation), which is the most complex workflow in the application.

The application follows accessibility principles well, with comprehensive audio feedback, intuitive gesture navigation, well-implemented haptic responses, and thoughtful multi-modal interaction design. The core functionality is accessible and user-friendly, with clear navigation patterns and reliable system responses throughout the user experience.

VII. DISCUSSION

The expert evaluation findings show that VisionTag represents a good example of accessible mobile application design, successfully addressing the complex challenges of creating effective assistive technology for clothing management. The application achieves a good balance between comprehensive

functionality and intuitive accessibility, providing visually impaired users with useful tools for independent wardrobe management.

A. Strengths and Design Success

VisionTag does well in implementing accessibility-first design principles throughout its interface. The application's gesture-based navigation system provides an intuitive alternative to traditional visual interfaces, while the comprehensive text-to-speech integration ensures users receive complete information about their interactions and options. The haptic feedback system adds important tactile confirmation that helps user confidence during navigation and task completion.

The application's two-mode structure (Home Mode and Retail Mode) creates clear functional boundaries that help users develop mental models for different types of interactions. The wardrobe management system in Home Mode provides effective tools for organizing and tracking clothing items, while the Retail Mode offers innovative QR code functionality that enables independent shopping and item documentation.

The onboarding experience deserves recognition for its effectiveness in teaching gesture vocabulary through practical application. By requiring users to perform actual gestures rather than simply describing them, the tutorial ensures users develop muscle memory and confidence before engaging with core application features.

B. Areas for Improvement

The few minor areas identified for potential improvement represent opportunities to improve the user experience. The suggested improvements focus on providing more detailed feedback during longer operations and further standardizing interaction patterns across different application contexts.

Better Progress Communication: While the current progress feedback system works well, adding more detailed progress announcements during QR generation and complex operations could provide additional user confidence. This improvement would build upon the foundation of system status communication.

Gesture Pattern Improvement: The minor variations in gesture implementation across different screens could be further standardized to create a more consistent interaction vocabulary. This represents an improvement of an already effective gesture system rather than a correction of significant inconsistencies.

Contextual Guidance Expansion: The existing help system could be expanded to provide more detailed guidance for complex workflows, particularly for users who are new to QR code technology or clothing management applications. This would improve the onboarding and help systems.

C. Future Work

This expert evaluation provides a foundation for further research on VisionTag's usability and accessibility. Future studies could conduct user testing with actual visually impaired users to validate our expert findings and identify

additional usability issues that may not be apparent through heuristic evaluation alone. Additionally, formal WCAG 2.1 compliance testing with certified accessibility auditors would provide definitive accessibility standards compliance verification. Comparative studies with other assistive technology applications in the clothing management domain could also help establish VisionTag's relative effectiveness and identify industry best practices for similar applications.

VIII. CONCLUSION

The expert evaluation confirms that VisionTag successfully achieves its goal of providing accessible, effective clothing management technology for visually impaired users. The application shows good attention to accessibility principles while delivering powerful functionality through an intuitive, well-designed interface.

The identified minor improvement opportunities represent potential improvements to an already successful implementation rather than corrections to significant usability problems. VisionTag stands as a good example of how assistive technology can provide effective functionality while maintaining accessibility and ease of use.

The application's innovative combination of QR code technology with comprehensive accessibility features creates a valuable solution that empowers users to manage their wardrobes independently and confidently. With its strong foundation in accessibility best practices and effective technical implementation, VisionTag represents a significant contribution to assistive technology in the fashion and personal care domain.

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